

# Bar Jacobi

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## Education

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**University of Washington** 2026 – Present  
*Ph.D. in Computer Science* *Seattle, WA*

**Tel Aviv University** 2024 – 2026  
*M.Sc. in Bioinformatics* *Tel Aviv, Israel*  
Advisor: Prof. Adi Stern  
Thesis: *Function prediction of viral proteins using language models*

**Technion – Israel Institute of Technology** 2017 – 2021  
*B.Sc. in Computer Science* *Haifa, Israel*

## Research Experience

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**University of Washington** 2026 – Present  
*Ruohola-Baker Lab, Department of Biochemistry* *Seattle, WA*

- Using single-cell data and foundation models to study how healthy cells transition toward aged states and identify changes that may be experimentally reversible.
- Working closely with experimental biologists to develop testable computational predictions, in collaboration with the Institute for Protein Design.

**Tel Aviv University** 2024 – 2026  
*Stern Lab* *Tel Aviv, Israel*

### *Viral protein function prediction*

- Developed models for predicting viral protein function from protein language-model representations of sequence and structure.
- Designed family-aware evaluation strategies to test whether models generalize to evolutionarily distant viruses, revealing that conventional data splits substantially overestimate performance.
- Compared sequence- and structure-based representations and investigated how predictive performance changes with evolutionary distance and biological function.

### *Within-host evolution of SARS-CoV-2*

- Studied intra-host viral diversity in chronically infected SARS-CoV-2 patients and acute-infection controls.
- Developed a replicate-concordance approach to distinguish genuine low-frequency variants from sequencing noise.
- Contributed to analyses of within-host evolution, resulting in a publication in *Virus Evolution*.

## Publications

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**Ben Zvi, A., Rutsinsky, N., Jacobi, B., et al. (2025).**

*Diverse patterns of intra-host genetic diversity in chronically infected SARS-CoV-2 patients.*

*Virus Evolution.* doi:10.1093/ve/veaf047

## Industry Experience

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**Meta (Facebook)**

2021 – 2024

*Production Engineer, Facebook Lite*

*Tel Aviv, Israel*

- Led infrastructure reliability work for Facebook Lite, serving more than 500 million monthly active users.
- Developed and maintained large-scale data pipelines, distributed monitoring systems, and reliability automation.

## Teaching Experience

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**Tel Aviv University**

2024 – 2025

*Teaching Assistant, Python Programming for Engineers*

**Technion – Israel Institute of Technology**

2021

*Teaching Assistant, Advanced Operating Systems*

**she codes;**

2016 – 2017, 2022

*Volunteer Instructor, Java Programming*

## Fellowships & Awards

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**Excellence in Research Award, Tel Aviv University**

2026

Faculty-wide annual research award to 30 outstanding graduate students in the Faculty of Life Sciences (500+ graduate students).

**Warren Alpert Computational Biology and AI Network Fellowship**

2025 – 2026

Selective fellowship in computational biology and AI through the Computational Genomics Summer Institute at UCLA.

**Edmond J. Safra Center for Bioinformatics Fellowship**

2024 – 2026

Graduate fellowship supporting training in bioinformatics at Tel Aviv University.

**Dean's Excellence Award, Technion – Israel Institute of Technology**

2019 – 2020

Semester-based academic distinction awarded to approximately the top 15% of Computer Science students (approximately 180–200 recipients per semester).

## **Presentations & Conferences**

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**Edmond J. Safra Center for Bioinformatics Annual Retreat** Tel Aviv, 2026  
Poster presentation; second place in the poster competition.

**Computational Genomics Summer Institute (CGSI)** UCLA, 2025  
Trainee and presenter.

**SBCR Conference: Viruses Across Domains of Life** Tel Aviv, 2025  
Poster presentation.

**Israeli Conference on Hardware and Side-Channel Attacks** 2021  
Competition finalist.

## **Leadership & Service**

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**Alumni Speaker, Technion Career Week** 2025  
CV workshop for students.

**Student Representative, Technion Student Association** 2018 – 2020

## **Technical Skills**

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|-------------------------|------------------------------------------------|
| <b>Programming</b>      | Python, Bash, C++                              |
| <b>Machine Learning</b> | PyTorch, scikit-learn, protein language models |
| <b>Bioinformatics</b>   | NGS analysis, phylogenetics, MMseqs2, iVar     |
| <b>Data Analysis</b>    | pandas, NumPy, SQL, scientific visualization   |
| <b>Computing</b>        | Linux, Docker, Git, HPC environments           |